
CDR

VOVAN-T

CDR AVAYA

Installation, source
configuration, and
administration

Docker release 2.5

Communication Manager ·
Session Manager · Equinox ·
SBCE · Other

Contents

Contents	2
1. Product overview	4
Deployment model	4
Security boundary	4
Installation	5
Host requirements	5
Online installation	5
Offline installation	5
First login and validation	5
Source configuration	6
2. Communication Manager	6
Customized CDR	6
Unformatted CDR	6
CM parameters	6
Acceptance call set	6
3. Session Manager	7
4. Equinox Management	7
5. SBCE	8
6. Other	8
Using the interface	9
7. Call journal	9
Details	9
View and dictionaries	9
Administration	10
8. Users and audit	10
Emergency password reset	10
9. Services and diagnostics	10
Logs	11
Support Pack	11
Backup, updates, and recovery	12
10. Backup and Restore	12
11. Cumulative update 2.5.4	12
Recovery decision	12
Troubleshooting and acceptance	13
12. Diagnostic sequence	13
Useful commands	13

13. Production acceptance checklist

13

1. Product overview

CDR AVAYA receives call-detail records from several Avaya platforms, preserves source data as RAW, and presents normalized calls and conference events in one web interface.

Layer	Responsibility
Collection	TCP listener for CM plus scheduled file retrieval for SM, EQ, SBCE, and Other.
RAW storage	Original parsed records remain available for audit, troubleshooting, and recalculation.
Processing	Independent processors build Calls, UCID chains, conference events, and linked details.
Presentation	Vue interface, Node API, filters, dictionaries, export, diagnostics, and administration.
Persistence	PostgreSQL 16 in the dedicated cdr-db container.

Deployment model

- cdr-app contains the web application, listener, processors, file fetcher, load balancer, maintenance loop, and command-line tools.
- cdr-db contains PostgreSQL and the persistent avaya_cdr database.
- Host folders under /opt/cdr exchange backups, updates, logs, uploaded files, SSH state, and manifests with the container.
- CDR_User is a restricted host SFTP account used by systems that push files to CDR.

Release model

A clean installation always reports 2.5.0. Functional releases are cumulative 2.5.x update packages. Update 2.5.4 installs directly over clean 2.5.0.

Security boundary

CDR serves HTTP on port 8010. Use nginx, HAProxy, or another reverse proxy for TLS. Keep PostgreSQL bound to localhost and do not expose the Docker socket to the application container.

PART I

Installation

Deploy a clean and supportable Docker installation before configuring telephony sources.

Host requirements

- x86-64 Ubuntu Server 24.04 or Astra Linux 1.7.
- Docker Engine and Docker Compose v2.
- Root or sudo access.
- Recommended production starting point: 4 CPU cores, 8 GB RAM, and 40 GB free disk space.
- Static address, working time synchronization, and source-to-CDR network routes.

Online installation

```
wget https://github.com/vovan-T/CDR-AVAYA/releases/latest/download/install.sh
chmod +x install.sh
sudo ./install.sh
```

Choose online, accept or change Web/API port 8010, and enter the CDR_User password twice.

Offline installation

Run install.sh in offline mode on an Internet-connected workstation, then transfer the generated installer, image archive, and checksum to the isolated server.

```
sha256sum -c cdr-avaya-docker-images-2.5.0.tar.gz.sha256
chmod +x install-offline.sh
sudo ./install-offline.sh
```

First login and validation

```
cd /opt/cdr
sudo docker compose ps
curl -sS http://127.0.0.1:8010/api/health
sudo docker exec cdr-app /opt/cdr/bin/cdr-control.sh post-update-check 2.5.0
```

Open `http://SERVER_IP:8010` and sign in with `admin/admin`. The UI requires a new administrator password. Inspect demonstration ACD CM1 and listener port 5001 before creating production ACDs.

Do not configure sources during installation
The installer deploys CDR only. All ACD, source, field, dictionary, and view settings are maintained in the web interface.

PART II

Source configuration

Configure every Avaya source independently and verify new RAW records before interpreting Calls.

2. Communication Manager

Customized CDR

Create or open an ACD, select CM format Customized, and configure CM to send records to the ACD listener address and port. The field list in CDR must match the CM layout exactly, including field widths.

Required field	Purpose
date / time	Call start date and time.
calling-num / dialed-num	Originating and destination numbers.
sec-dur	Duration in seconds.
cond-code	Avaya condition code used while walking the call chain.
in-trk-code	Incoming trunk access code and route context.
ucid / seq-num	Chain identity and record order.
code-dial / code-used	Dialed and selected route codes.

Mandatory profile values are shown in red and read-only. Additional fields may be inserted when they are configured on CM with the exact expected width.

Unformatted CDR

Unformatted mode does not contain UCID and is not suitable for reliable SIP call-chain reconstruction. CDR therefore presents the first and last endpoint available in the record sequence and treats non-local routes conservatively.

Enhanced unformatted

Use Enhanced Formats = n. Enhanced unformatted parsing is intentionally blocked until its behavior is validated without a simultaneous Customized stream.

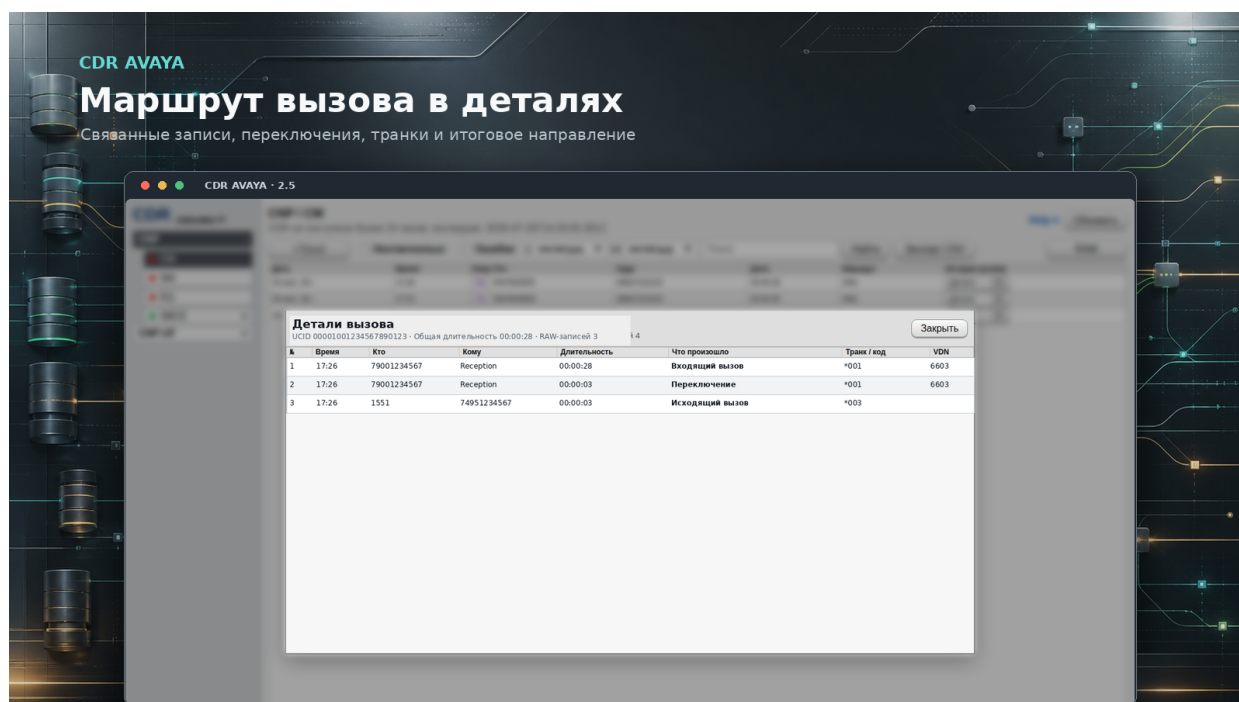
CM parameters

- Port must be present and unique among active listeners.
- Use the red read-only values as the required Communication Manager profile.
- Debug enables detailed RAW diagnostics and should be disabled during normal operation.
- Split/attendant-related values affect CM output and must be tested against the selected layout.
- View controls displayed columns only; it does not change processing or RAW storage.

Acceptance call set

- local answered and unanswered calls;
- incoming and outgoing trunk calls;
- VDN and announcement routing;
- transfer, consultation, forwarding, and conference;

- transit path through CM and Session Manager;
- the same scenarios in RAW and Calls views, with Details checked by UCID.



CM call chain: the journal keeps the first caller and final destination while Details exposes every linked RAW stage.

3. Session Manager

CDR supports the extended Session Manager XML format. Configure an SSH/SFTP account on SM with read access to the documented CDR export directory. The remote directory is fixed by the Avaya SM configuration and is not a free-form staging location.

- 1 Verify a normal SSH/SFTP login from the CDR host. Host-key confirmation is automated by the collector during the first standard connection.
- 2 In the SM source settings, enter server label, host, SSH port, username, password, and the hourly/minute download schedule.
- 3 Save the source and run Diagnostics. Confirm that XML files appear in RAW and that Calls contains the expected server label.
- 4 Open Details to inspect all RAW records linked by the resulting call_key.

4. Equinox Management

Equinox Management may create files with permissions that prevent direct collection. A root-side staging script copies every cdr*.xml file into the CDR_User home area with readable ownership. Files are retained and synchronized again when their content changes.

- 1 Copy the supplied cdr_collector.sh to the CDR_User home directory on Equinox Management.
- 2 Run the script as root and wait for the final OK message.
- 3 Verify source and staging counts in cdr_collector.log.
- 4 Install the root cron entry at minute 5 of every hour; configure CDR retrieval at minute 10.
- 5 In the EQ source settings, point CDR at the staged CDR_User directory and verify conference events and participants.

Changing files

Synchronization is content-aware. A new file is imported; a changed file is fetched again and compared by hash so newly appended records can be processed.

5. SBCE

The installer creates CDR_User and /opt/cdr/data/sbce/1. Configure the SBCE CDR adjunct to upload into path 1 relative to the restricted SFTP root.

Setting	Value
Host	CDR server address
Port	22
Username	CDR_User
Path	1 for the first instance
Schedule	Configured by the SBCE CDR adjunct
Log	/archive/log/icu/cdr_sftp.log

To force a test on SBCE, run /usr/local/ipcs/icu/scripts/sftp_cdr.py and inspect the adjunct log. In CDR, verify the instance RAW view, Calls, linked Details, Routing Profile, and Server Flow.

6. Other

Other is a user-defined collector for non-Avaya or custom data. Choose TCP or file retrieval, then Flat or XML parsing. Define record boundaries and fields before enabling the source.

- Flat line mode maps delimited or fixed records into configured columns.
- XML mode uses configured record and value paths.
- SCP/SFTP sources use the shared server schedule and preserve downloaded files.
- Start with an inactive source, use a small sample, and inspect RAW before enabling continuous collection.

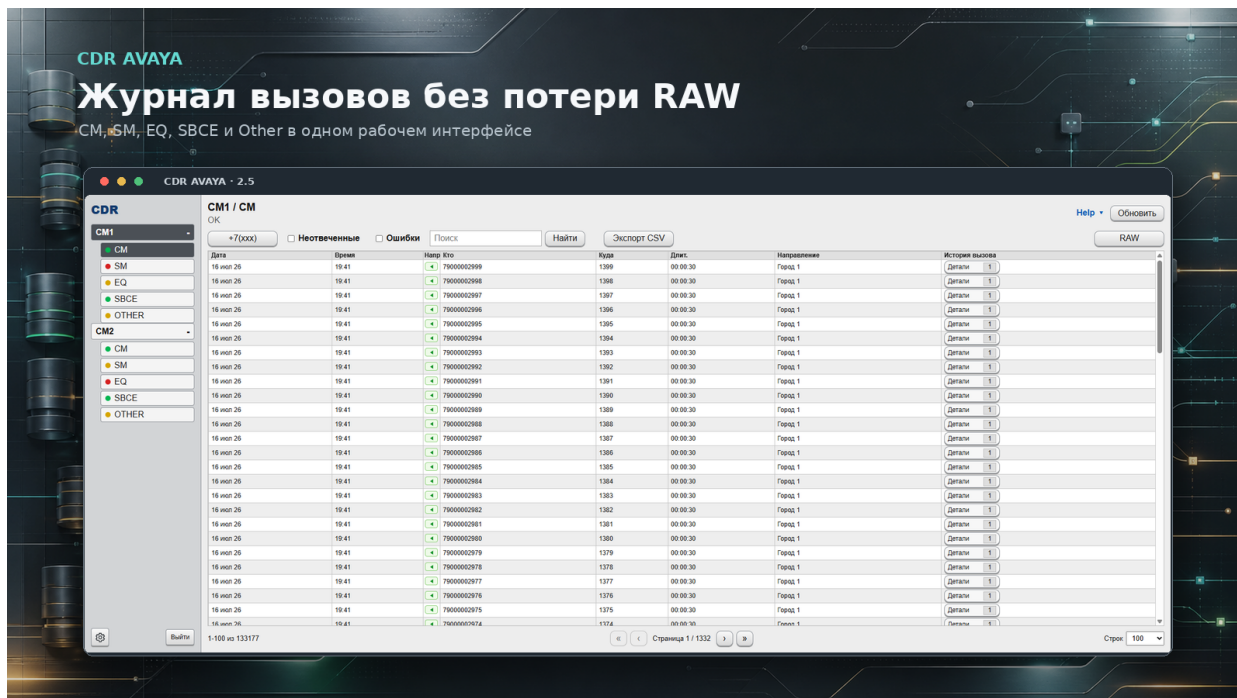
PART III

Using the interface

The journal is designed for repeated operational analysis rather than one-time reports.

7. Call journal

- Select ACD and source in the left tree.
- Use date, number, and text search to narrow the current view.
- Unanswered and Errors checkboxes add those states to the normal result set.
- The right-click filter shows only the chosen value; All clears that exact filter.
- RAW switches between normalized Calls and preserved source records.
- Export CSV uses the active filters and displayed column set.



The CM journal with source tree, additive status filters, date range, search, export, and RAW mode.

Details

Details is not another call result. It is the ordered evidence used to build the selected row. CM Details follows UCID and sequence. SM/SBCE Details follows the stored call_key and linked RAW identifiers.

View and dictionaries

View defines visible columns, labels, ordering, and formatting rules. Dictionaries translate technical values such as trunk codes, VDNs, numbers, and routes into readable labels. Neither feature deletes or rewrites RAW data.

Number formatting

Formatting rules are display-only. Place more specific patterns before general patterns and test 10- and 11-digit examples, including city codes with different lengths.

PART IV

Administration

Use the System area for users, audit, services, storage, updates, diagnostics, and interface defaults.

8. Users and audit

Administrators create users and assign per-ACD read or edit access. At least one administrator must remain. The audit journal records authentication, configuration changes, ACD operations, service control, Backup/Restore, and updates without storing passwords or secrets.

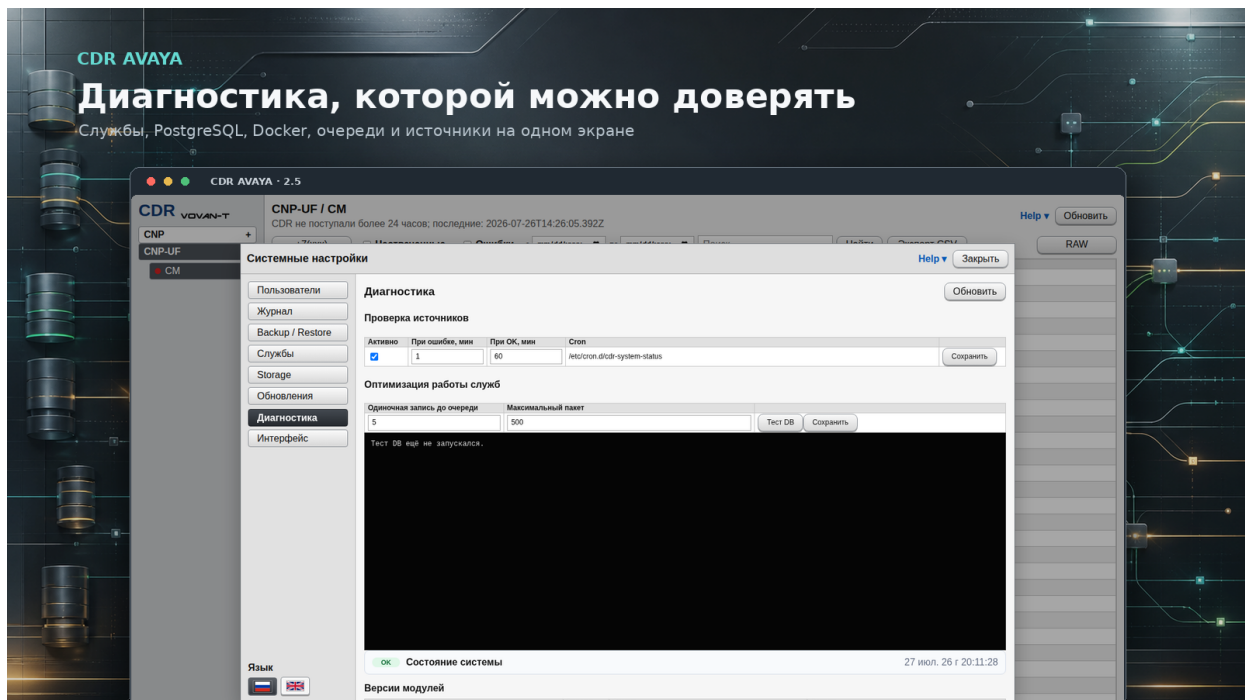
Emergency password reset

```
sudo docker exec -it cdr-app /opt/cdr/venv/bin/python /opt/cdr/bin/cdr-reset-password.py admin
```

9. Services and diagnostics

Start, Stop, and Restart target the selected supervised process. A normal API restart does not restart listeners or processors and does not recreate the container.

```
sudo docker exec cdr-app /opt/cdr/bin/cdr-control.sh docker-diagnostics
sudo docker exec cdr-app /opt/cdr/bin/cdr-control.sh post-update-check 2.5.4
```



Expanded service diagnostics show every supervised child, runtime checks, API/DB health, and recent supervisor events.

Logs

File	Purpose
logs/api.log	Web API, authentication, settings, Backup/Restore, and update requests.
logs/listener.log	CM TCP listener and RAW ingestion.
logs/processor*.log	Per-source RAW-to-Calls processing.
logs/sm-fetch.log	Session Manager file retrieval.
logs/eq-fetch.log	Equinox file retrieval.
logs/sbce-fetch.log	SBCE folder scan and import.
logs/update.log	Update, rollback, commit, and post-update actions.
logs/backup.log	Backup/Restore stages and critical failures.

Support Pack

Use System -> Diagnostics -> Support Pack -> Create. The archive contains diagnostics, logs, version data, safe SQL metadata, and configuration evidence required by support. It excludes passwords and application session secrets.

PART V

Backup, updates, and recovery

Keep application release management separate from business-data recovery.

10. Backup and Restore

Standard Backup creates a full PostgreSQL custom-format dump plus a Backup manifest. It does not include application code, previous backup archives, logs, or temporary files.

```
sudo docker exec cdr-app /opt/cdr/bin/cdr-backup.sh --database
```

```
sudo docker exec cdr-app /opt/cdr/bin/cdr-backup.sh --restore /opt/cdr/backup/cdr-db-HOST-YYYYMMDD-HHMMSS.tar.gz
```

Exact DB version

Standard Restore proceeds only when backup_db_version equals system_db_version. A different version requires Migration Tool. Restore checks this before stopping processors or changing PostgreSQL.

During Restore, the current database is renamed, a clean database is created, and pg_restore is validated. If any critical step fails, the incomplete database is removed and the previous database name is restored.

11. Cumulative update 2.5.4

```
sudo cp update-2.5.4.tar.gz update-2.5.4.tar.gz.sha256 /opt/cdr/update/
cd /opt/cdr/update
sudo sha256sum -c update-2.5.4.tar.gz.sha256
```

Apply the package in System -> Updates. The package is cumulative and supports clean Docker 2.5.0 directly.

Action	Result
Apply	Validates compatibility and SHA, creates rollback, installs files, runs DB migrations, and restarts required child processes.
Rollback	Restores previous files, manifest, database copy, and CDR_User state.
Commit	Accepts the update and removes rollback files and the temporary rollback database.

Recovery decision

- If the update completed and the application starts, use Rollback before Commit.
- If only business data or settings are wrong, restore the pre-update same-version database backup.
- If application files are incomplete and rollback is unavailable, install clean Docker 2.5.0, apply the required cumulative update, then restore the matching DB backup.
- Never copy PostgreSQL data-directory files while the database is running.

PART VI

Troubleshooting and acceptance

Separate source connectivity, collection, RAW parsing, processing, and UI presentation while diagnosing a problem.

12. Diagnostic sequence

- 1 Confirm cdr-app and cdr-db are healthy and API health returns status=ok, db=ok.
- 2 Run expanded Docker diagnostics and inspect recent supervisor events.
- 3 Confirm source connectivity with the same host, port, username, and path configured in CDR.
- 4 Check the source fetch or listener log for the exact command and answer.
- 5 Confirm that RAW count increases. If RAW is empty, do not debug the processor or UI.
- 6 If RAW grows but Calls does not, inspect parser errors, pending stat=false rows, processor log, and source format.
- 7 If Calls exists but the UI view is confusing, clear right-click filters, inspect View columns, open RAW and Details, and compare the same call_key.
- 8 Create a Support Pack before manual cleanup or recalculation.

Useful commands

```
cd /opt/cdr
sudo docker compose ps
curl -sS http://127.0.0.1:8010/api/health
sudo docker logs --tail 120 cdr-app
sudo docker logs --tail 120 cdr-db
sudo docker exec cdr-app /opt/cdr/bin/cdr-control.sh docker-diagnostics
```

13. Production acceptance checklist

- Docker containers are healthy after reboot and cdr-app RestartCount remains zero.
- Administrator password is changed and at least one recovery procedure is documented.
- HTTPS reverse proxy or trusted-network restriction protects port 8010.
- Every active source produces new RAW without parser errors.
- Representative local, incoming, outgoing, transfer, forwarding, conference, and transit calls are verified where supported.
- SM/SBCE Details and SBCE Routing Profile are verified on live records.
- Backup is created, checksum is present, and a test Restore is completed on a non-production system.
- Update 2.5.4 is committed only after health, diagnostics, source queues, and user access are checked.
- Support Pack can be generated and downloaded.

Operational principle

Preserve RAW first. Processing rules and presentation can be recalculated later; missing source records cannot be reconstructed.